A Note on the Solutions: This solution sheet is intended for you verify correctness of your work during the tutorial. The solutions here do not necessarily meet the standards for exposition expected on assignment submissions.

Recall the following theorems from Week 5:

Theorem (Divergence Criteria Theorem)**.** *Let (f_n) be a sequence. If any of the following are true, then (f_n) diverges.*

1. *(f_n) is unbounded*
2. *(f_n) has two subsequences each of which converge to a different value*
3. *(f_n) has a divergent subsequence*

1. **Divergent Sequences**

   For each sequence below, use the Divergence Criteria Theorem to determine if sequence diverges. Caution! There may be some sequences below for which the Divergence Criteria Theorem gives us no information.

   (a) $(a_n) = (\sqrt{n})$.

   (b) (b_n) for

   $$b_n = \begin{cases} \frac{1}{n} & n = 3k \\ 1 & n = 3k+1 \\ -\frac{1}{2n} & n = 3k+2 \end{cases}$$

   (c) (c_n) for

   $$c_n = \begin{cases} \frac{1}{n} & n = 3k \\ 0 & n = 3k+1 \\ -\frac{1}{2n} & n = 3k+2 \end{cases}$$

   (d) $(d_n) = (n - 10^{100}\sqrt{n})$

   Solutions:

   (a) By the Archimedean Principle, for all $r \in \mathbb{R}$ there exists $n \in \mathbb{N}^+$ such that $n > r^2$. Hence $\sqrt{n} > r$. Therefore, the sequence $\left(\sqrt{n}\right)$ is unbounded and diverges.

   (b) This sequence has two subsequences that converge to different limits. The subsequence $(b_1, b_4, b_7, \dots)$ converges to 1. The subsequence $(b_2, b_5, b_8, \dots)$ converges to 0.

   (c) The subsequence divergence test tells us nothing because this sequence converges.

   (d) Note that $d_n = \sqrt{n}\left(\sqrt{n} - 10^{100}\right)$. By the Archimedean Principle, for all $r \in \mathbb{R}$ there exists a natural number m such that $\sqrt{m} > 10^{100} + 1$. Hence for all $n > m$, $d_n > \sqrt{n}$. Applying the Archimedean Principle again, we see that for all $r \in \mathbb{R}$ there exists $n \in \mathbb{N}^+$ (even larger than the previous one) such that

   $$d_n = n - 10^{100}\sqrt{n} = \sqrt{n}\left(\sqrt{n} - 10^{100}\right) > r.$$

   Therefore the sequence is unbounded and so diverges.

2. **A Squeeze Theorem for Sequences**

   In this exercise our goal is to prove the following theorem.

   Theorem. *Let $(f_n), (g_n)$ and (h_n) be sequences. If $f_n \leq g_n \leq h_n$ for all $n \in \mathbb{N}^+$ and $\lim_{n\to\infty} f_n = \lim_{n\to\infty} h_n = L$, then $\lim_{n\to\infty} g_n = L$*

   (a) What does it mean when we write $\lim_{n\to\infty} f_n = L$. Use the notation M_f instead of M.

   (b) What does it mean when we write $\lim_{n\to\infty} h_n = L$. Use the notation M_h instead of M.

   (c) Let $M = \max\{M_f, M_h\}$, let $\epsilon > 0$ and let $n > M$. Briefly explain how you know $L - \epsilon < f_n < g_n < h_n < L + \epsilon$.

   (d) Manipulate the expression $L - \epsilon < g_n < L + \epsilon$ to verify $g_n \to L$.

Solutions:

(a) For every $\epsilon > 0$ there exists $M_f \in \mathbb{N}^+$ so that $|f_n - L| < \epsilon$ whenever $n > M_f$.

(b) For every $\epsilon > 0$ there exists $M_h \in \mathbb{N}^+$ so that $|h_n - L| < \epsilon$ whenever $n > M_h$.

(c) Since $M \geq M_f$ we have $|f_n - L| < \epsilon$. Therefore $f_n - L > -\epsilon$ and so $f_n > L - \epsilon$.

Since $M \geq M_h$ we have $|h_n - L| < \epsilon$. Therefore $h_n - L < \epsilon$ and so $h_n < L + \epsilon$.

(d)

$$\begin{aligned} L - \epsilon < g_n &< L + \epsilon \\ -\epsilon < g_n - L &< \epsilon \\ |g_n - L| &< \epsilon \end{aligned}$$

3. **Sequences Approaching Sequences**

In this course, we are focusing on the $\epsilon - M$ notion of limit, where a sequence f_n may approach a specific real number L as $n \to \infty$. However, this approach doesn't let us compare two diverging sequences. For example, how can we compare the sequences (n^2), $(n^2 + 4n)$ and (2^n). All three of these sequences diverge by virtue of being unbounded. However the third sequence grows at a rate much larger than the first two. The sequences (n^2), $(n^2 + 4n)$ seem to grow at about the same pace.

More precisely,

$$\lim_{n\to\infty} \frac{n^2 + 4n}{n^2} = 1.$$

In general, for $f_n, g_n \neq 0$, we say f_n <u>asymptotically approaches</u> g_n when

$$\lim_{n\to\infty} g_n/f_n = 1.$$

When f_n <u>asymptotically approaches</u> g_n we write $f_n \sim g_n$.

(a) Using the definition of asymptotically approaches show $(n^3 + 1)$ asymptotically approaches (n^3).

(b) Using the definition of asymptotically approaches show (n^2) does not asymptotically approach (2^n).

(c) Show that if $f_n \sim g_n$, then $g_n \sim f_n$. (Caution! There is no guarantee that either of (f_n) or (g_n) converge!)

If you are a student with a background in Computer Science, you may be familiar with the asymptotic run-times of algorithms and big-oh notation. We say <u>f is big-oh of g</u> when the sequence $\frac{f_n}{g_n}$ is bounded. When f is big-oh of g we write $f(n) \in O(g(n))$. An algorithm runs in *polynomial time* when the runtime of the algorithm is big-oh of a polynomial.

Solutions:

(a) Show $\lim_{n\to\infty} \frac{n^3}{n^3+1} = 1$. This can be easily done using the Algebra of Limits Theorem, or by applying the definition of convergence directly.

(b) Show that the sequence $\frac{2^n}{n^2}$ diverges by virtue of being unbounded. This requires some work. For instance, we can use induction to show that $2^n \geq n^3$ for all integers $n \geq 10$. Hence $\frac{2^n}{n^2} \geq n$ for all integers $n \geq 10$, and so the sequence $\frac{2^n}{n^2}$ is unbounded by an application of the Archimedean Principle.

(c) Let $h_n = (1, 1, 1, 1, \dots)$ and apply the last part of the Algebra of Limits Theorem to h_n and $\frac{g_n}{f_n}$